Mugilan Mariappan

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Education

2018 - present **Ph.D. in Computing Science**, *Simon Fraser University*, Canada.
Advisor: Dr. Keval Vora

2010 - 2014 **B.E. in Computer Science**, *Coimbatore Institute of Technology*, India.
GPA: 8.5/10

Research Interests

Streaming Graph Processing, Incremental Computing, Runtime optimizations, Graph Mining, Graph Databases, Graph Query Languages

Publications

GRADES-NDA 2022 **Anti-Vertex for Neighborhood Constraints in Subgraph Queries**
Kasra Jamshidi, Mugilan Mariappan, and Keval Vora
ACM SIGMOD Workshop on GRADES-NDA, 2022
[paper]

EuroSys 2021 **DZiG: Sparsity-Aware Incremental Processing of Streaming Graphs**
Mugilan Mariappan, Joanna Che, and Keval Vora
European Conference on Computer Systems (EuroSys), 2021
[video, paper, slides, code]

EuroSys 2019 **GraphBolt: Dependency-Driven Synchronous Processing of Streaming Graphs**
Mugilan Mariappan and Keval Vora
European Conference on Computer Systems (EuroSys), pages 25:1-25:16, 2019
[video, paper, slides, code]

Work Experience

May 2018 - present **Research Assistant**, *Simon Fraser University*, Canada.
Parallel & Distributed Computing Lab

May 2022 - Aug 2022 **Teaching Assistant**, *Simon Fraser University*, Canada.
CMPT 300 Operating Systems

Sep 2019 - Dec 2020 **Teaching Assistant**, *Simon Fraser University*, Canada.
CMPT 431 Distributed Systems

2014 - 2017 **Software Engineer**, *athenaHealth*, India.

Projects

2022 - present **Scalable Systems for GNNs**

- Researching on algorithmic and system optimizations for large-scale GNN systems.
- C++, Cuda, OpenMP.

2021 - 2022 **Anti-Vertices for Expressing Patterns in Graph Query Languages**

- Identified expressivity concerns for describing structural constraints in Graph Query Languages like Cypher.
- Formalize the matching semantics and develop ***prototype extensions to the Cypher grammar***.

2019 - 2020 **DZiG**

- An incremental processing technique which exploits the sparsity inherent in graph computations.
- ***Pushes the effectiveness of dependency driven computations*** to process over 10M simultaneous graph mutations on a single machine.
- Code open sourced as part of the GraphBolt project.
- C++, Cilk Plus, mimalloc.

2019-2020 **OLFIN**

- A system for performing efficient serializable graph mutation for multi-source input streams.
- Designed linearizable lockless graph mutation operations.
- Designed a light-weight ordering protocol to support bounded relaxed ordering to strict serializability of operations.
- C++, Cilk Plus, mimalloc.

2018 - 2019 **GraphBolt**

- Employs dependency-driven incremental processing to quickly get the updated results for dynamic graphs.
- ***Orders of magnitude faster*** at providing the results for the latest version of the graph while also ***guaranteeing correctness***.
- Code open sourced to github.
- C++, Cilk Plus.

2020 **NUMA-Aware Serializable Graph Mutations for Concurrent Streams**

- *CMPT 984 ST-Database Systems* course project.
- Created a NUMA-aware graph data structure and ordering protocol for processing graph updates from multiple streams on NUMA machines.
- C++, Cilk Plus.

2019 **GPU-based Graph Framework with Activation and Delta Propagation**

- *CMPT 886 ST-Parallel and Distributed Computing* course project.
- Incorporated vertex activation in GPU-based graph processing to eliminate redundant computation without losing warp efficiency.
- C++, Cilk Plus, Cuda.

2019 **Incremental Computation in Streaming Graphs - Under the Hood**

- *CMPT 886 ST-Large Scale Graph Processing* course project.
- Investigated the incremental refinement in GraphBolt and created a more efficient recursive refinement algorithm.
- C++, Cilk Plus.

2019 **Evaluating Locking/Synchronization Primitives**

- *CMPT 740 Database Systems* course project.
- Implement and evaluate the performance of spin locks, ticket locks and MCS locks on different workloads.
- C++, C++ threads, perf, Intel VTune.

2014 - 2017 **Projects at athenaHealth**

- Rearchitected the messaging module of athenaText, a secure messaging application for health providers. This improved performance and reduced crashes by over 50%.
- Development of Search in Epocrates, a top iOS app used by over 50% of physicians in USA.
- Objective C, Java, Apache Solr, Apache Spring.

Awards

2022	Helmut & Hugo Eppich Family Grad Scholarship
2022	CMPT Graduate Fellowship, Simon Fraser University
2022	Graduate Fellowship, Simon Fraser University
2021	CMPT Graduate Fellowship, Simon Fraser University
2020	Graduate Fellowship, Simon Fraser University
2019	Best Poster Award *(with Joanna Che)*, CS Research Day, Simon Fraser University
2019	CMPT Graduate Fellowship, Simon Fraser University
2018	Graduate Fellowship, Simon Fraser University
2018	Special Grad Entrance Scholarship, Simon Fraser University
2014 – 2017	Awarded Spotlight Award three times by athenaHealth

Other Activities

2022 and 2018-2019	CS Councillor - Graduate Student Society Simon Fraser University
2018-2019	Secretary - CS Graduate Student Association Simon Fraser University
2013-2014	Secretary - CS Association Coimbatore Institute of Technology

Mentoring

1. Joanna Che
   Undergrad and M.Sc., Simon Fraser University
2. Matthew Morrison
   Undergrad, Simon Fraser University
3. Durga Keerthi Mandarapu
   Intern, Simon Fraser University